GEM Cluster Timing

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Darklight Software Meeting

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What is the point of this presentation?

When there is a particle track, when do we detect the clusters in each GEM?

(relative to the trigger time)

- How do we define the time of a cluster?
- Can we distinguish “good” clusters from ”junk” clusters based on their timing relative to scintillator hits?

Previous work

Credit: Sophie Gelinas

Co-op students have looked into timing of GEM data in simulation.

- Sophie looked at how the ADC value of the “peak” channel of the cluster changes over time samples
- Andy looked at using the GEM as a time-projection chamber to reconstruct track angles

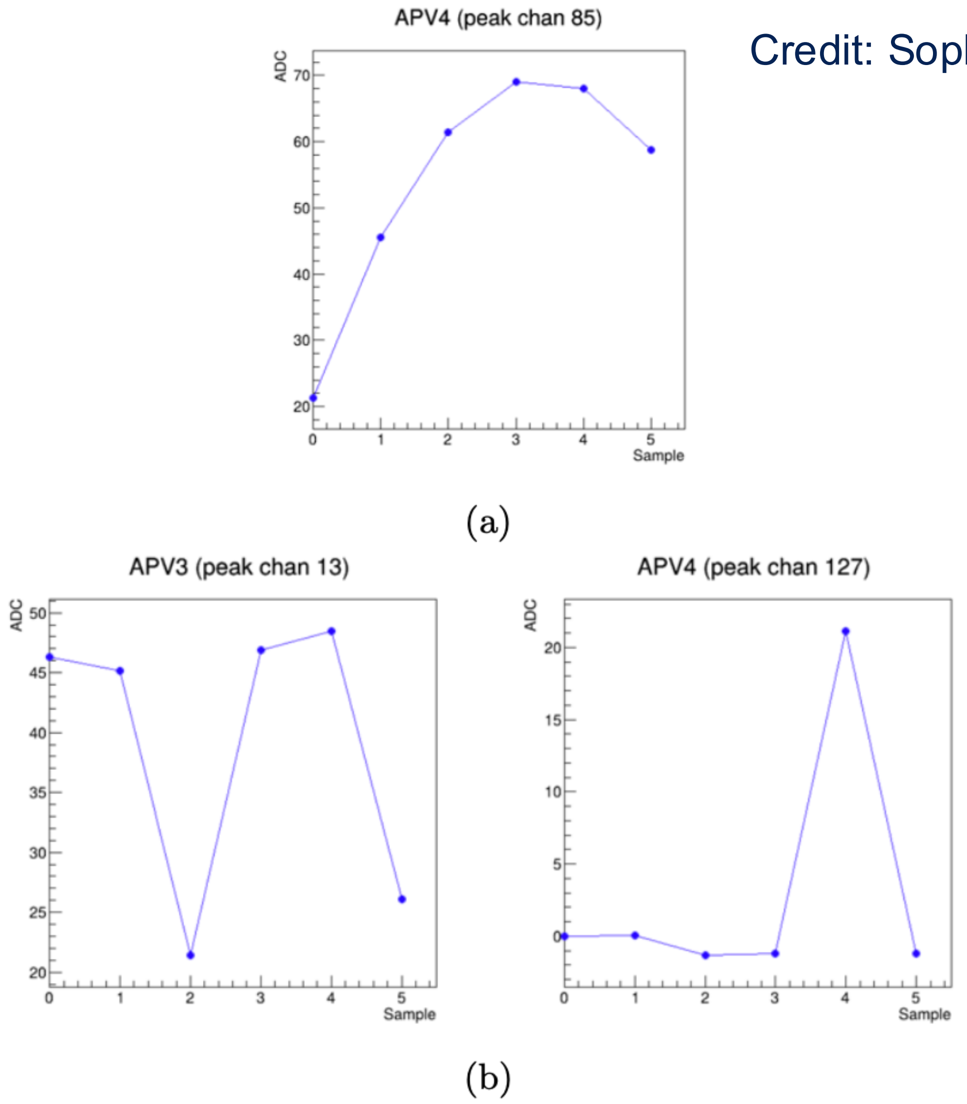


Figure 5: ADC of peak channel vs. sample number for a single APV. (a) shows an x-axis APV, and (b) shows two y-axis APVs.

Time of a GEM Cluster (First order)

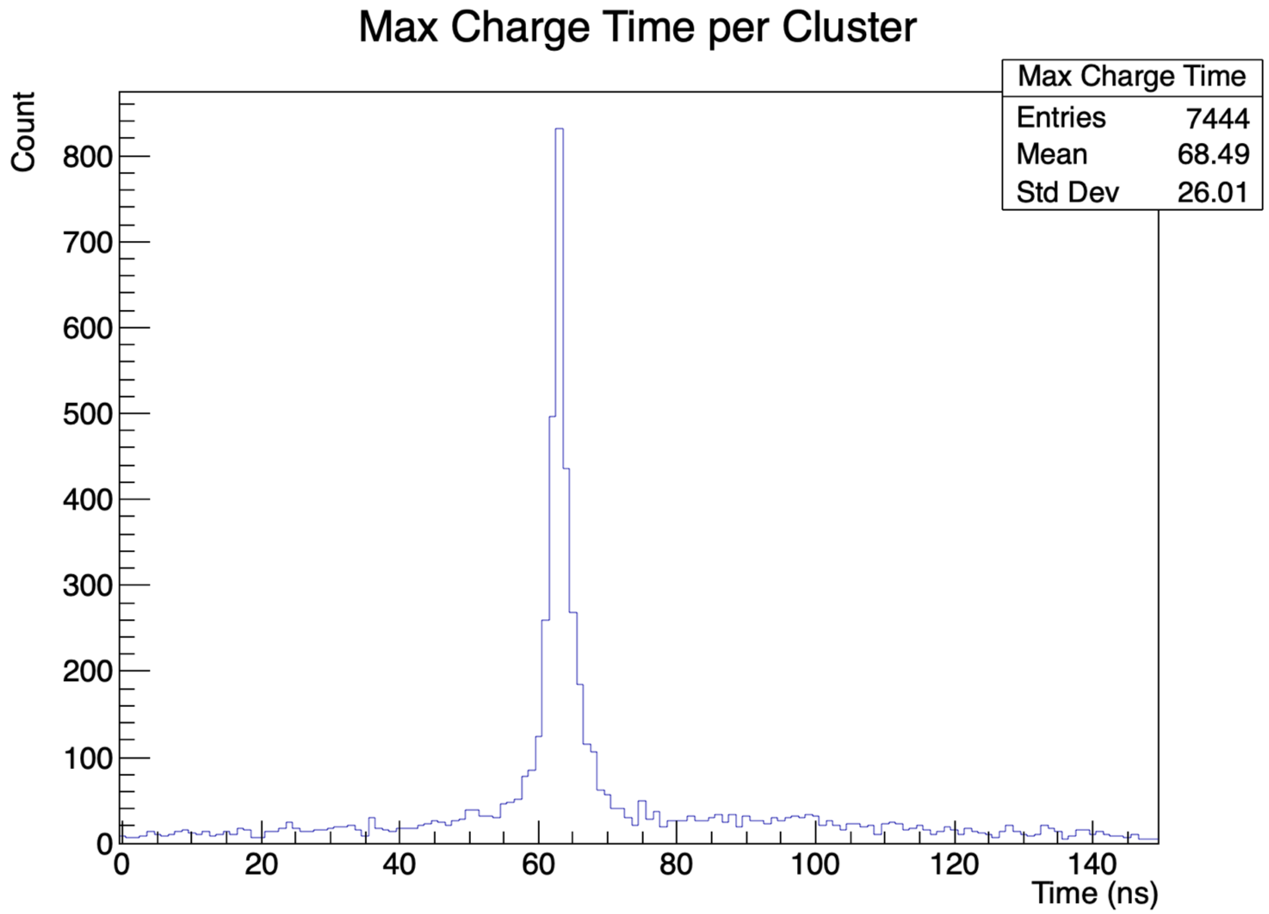
1. "Max Charge Time"
   - Charge ≡ sum of ADC values in all channels in cluster
   - How this is calculated:
     - Calculate charge of each sample
     - Calculate "Max Charge Time" as mean sample time weighted by charge of each sample

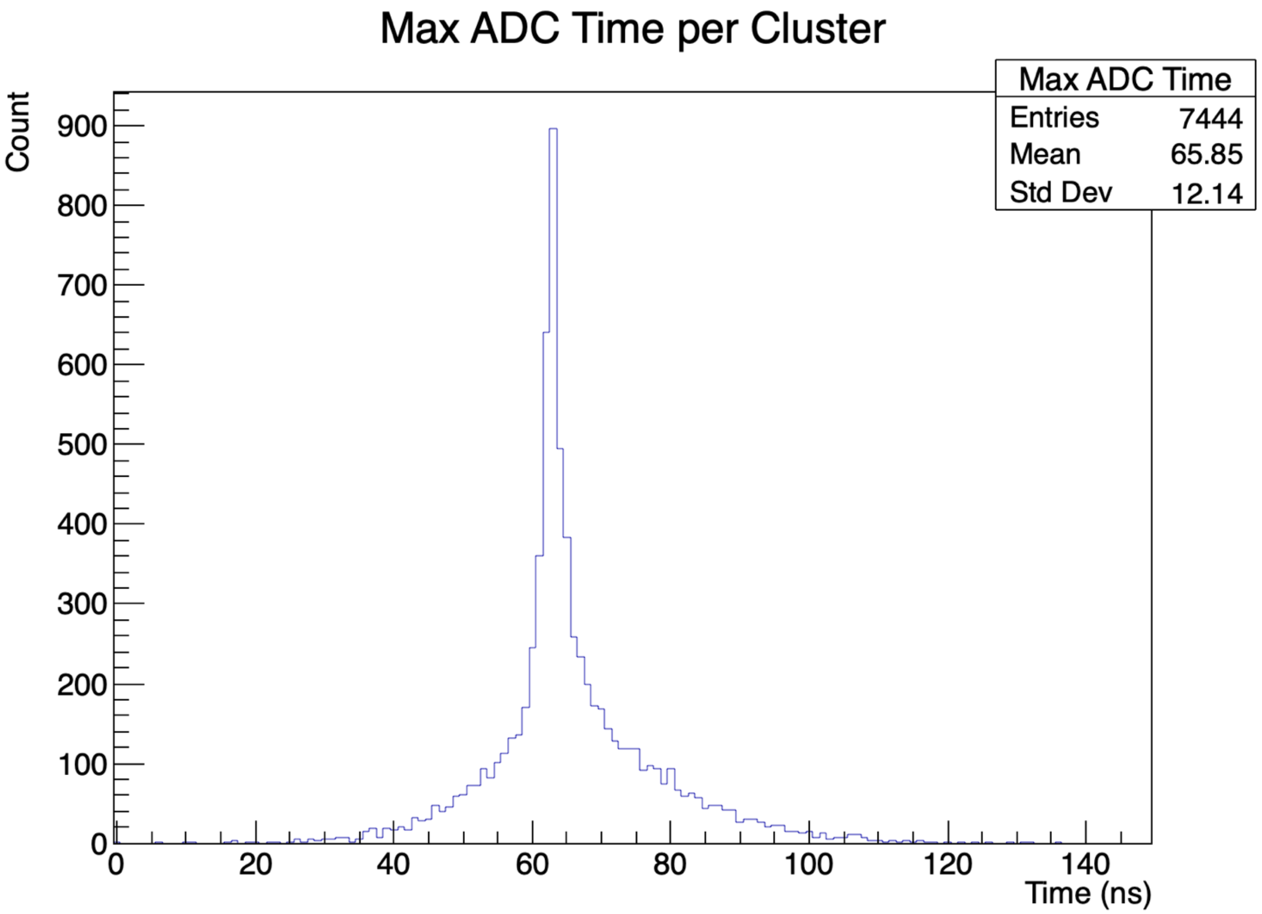
2. "Max ADC Time"
   - How this is calculated:
     - Find maximum ADC value in each time sample
     - Calculate "Max ADC Time" as mean sample time weighted by maximum ADC value in each sample

Cluster Time Distribution

Run 2679
Cosmic
Full Readout
100k Events

Left Top GEM – X axis cluster times





Next Steps

- Investigate correlation between first-order cluster time and trigger times in good cosmic ray tracks
- Implement fitting function to improve measure of cluster time

GEMini Cluster Finder

```cpp
if (false)
{ // Previous Method

    input_data data;
    int index = 0;
    for (int j = pos - 4; j <= pos + 4; j++)
    {
        data.x[index] = j - pos;
        data.y[index] = values[j];
        index++;
    }

    const u_int m = 9, n_par = 3; // Initializing a ton of cminpack stuff
    int maxfev, mode, nprint, info, nfev, ldfjac, ipvt[n_par];
    double ftol, xtol, gtol, epsfcn, factor, fnorm, par[n_par], diag[n_par], qtf[

    std::unique_ptr<double[]> fjac = std::make_unique<double[]>(m * n_par);
    std::unique_ptr<double[]> fvec = std::make_unique<double[]>(m);
    std::unique_ptr<double[]> wa4 = std::make_unique<double[]>(m);

    par[0] = values[(int)pos];
    par[1] = 0;
    par[2] = 0.7;
```

Gaussian fitting method

GEMini Cluster Finder

- Instead, the code decides if something is a cluster if its amplitude passes a threshold

```
                double sig;
                double med;
                GetMedianAndFilteredStd(pereventhisto, sig, med, 1.0);

                if (max - min > 5 * sig)
                {
                    // std::cout << "max: " << max << " min: " << min << " sig: " << sig <<
                    // std::cout << "max-min" << max-min << " 5*sig: " << 5*sig << std::end
```

- If anyone has information about this code, please let me know

GEMTrack Errors

After workfest, my code compiles but crashes when running GEMtrack

- Missing library files
- “terminating due to uncaught exception of type std::bad_alloc: std::bad_alloc”

Thank you

Left Bottom GEM X

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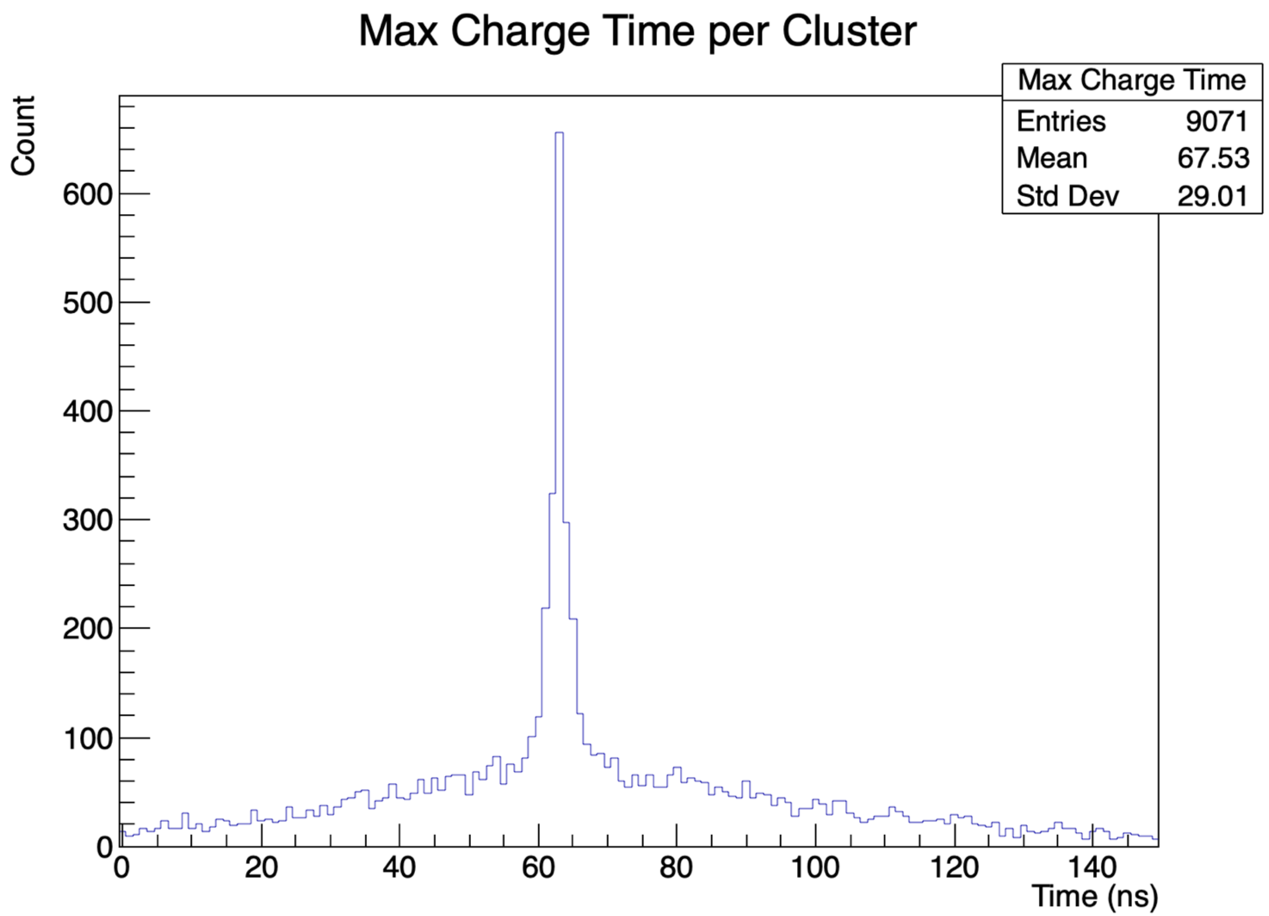

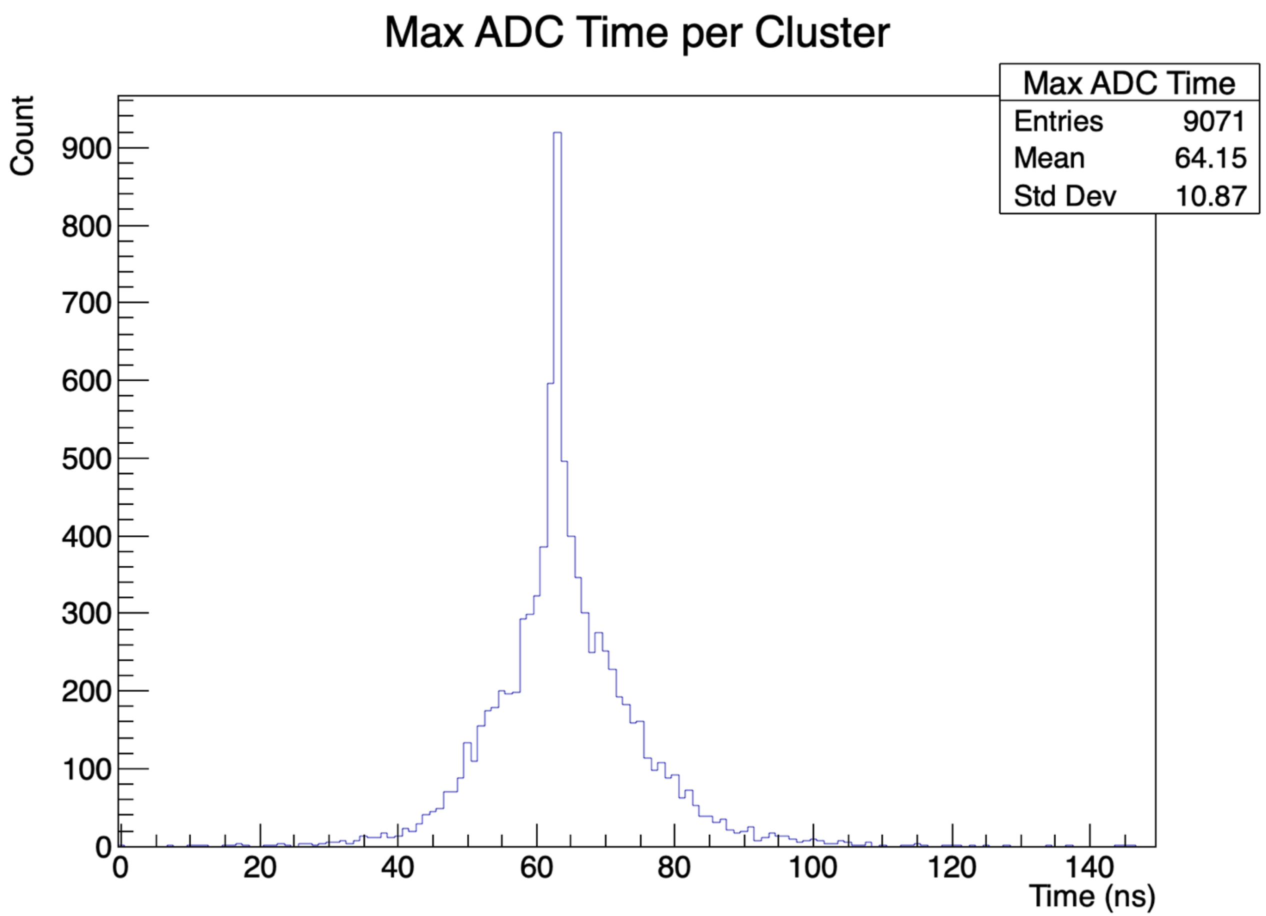

Left Top GEM X

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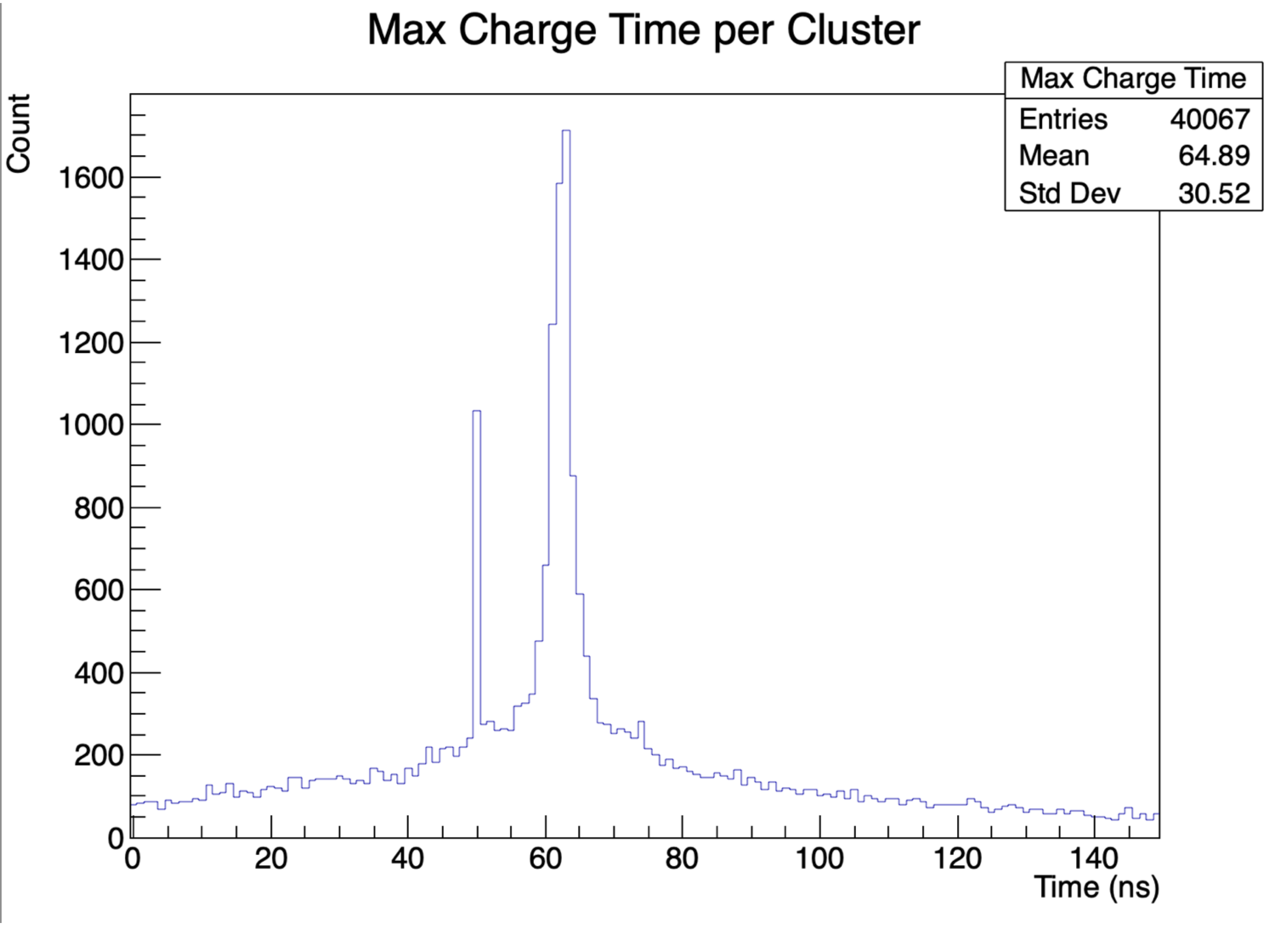

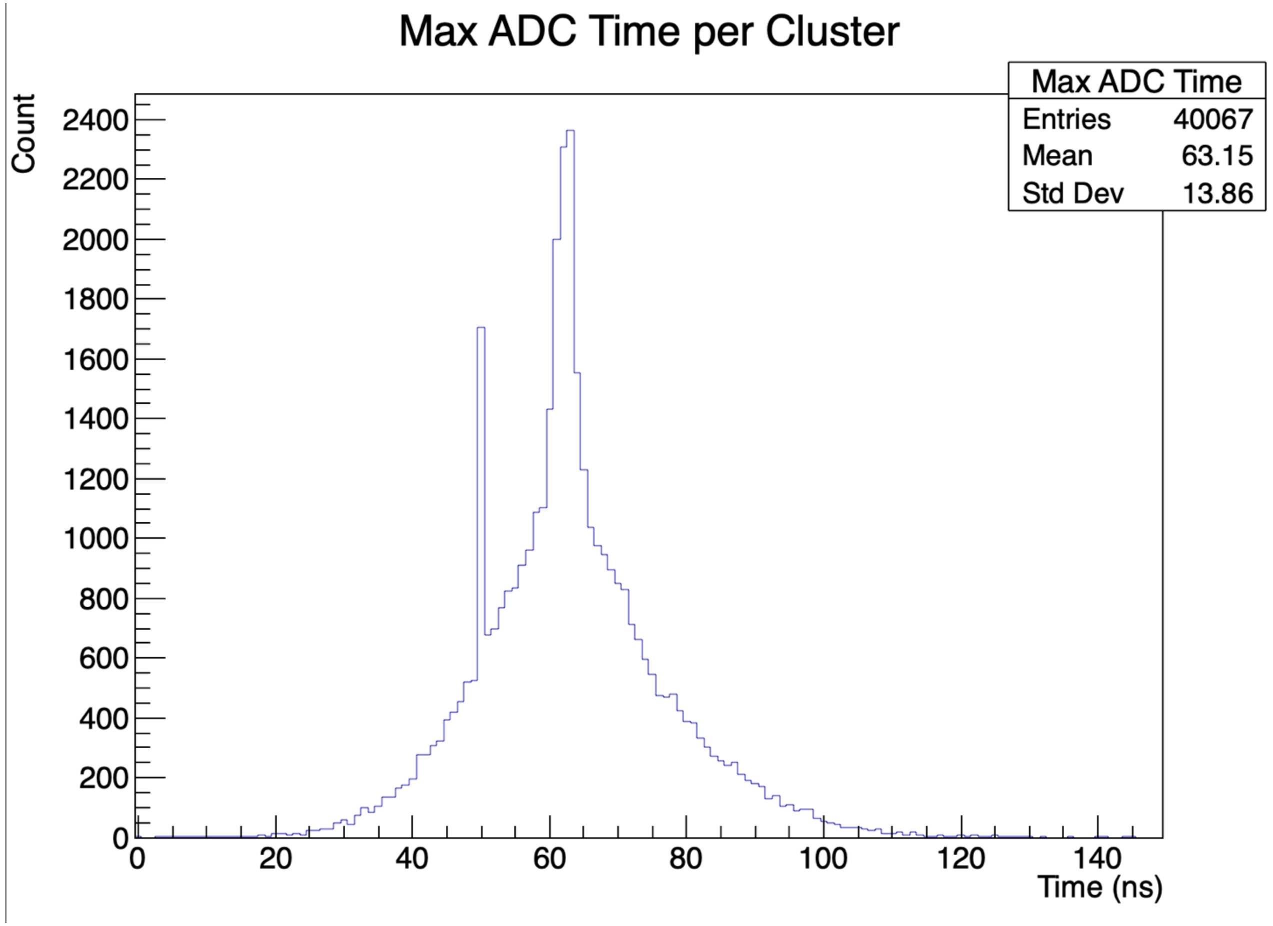

Left Top GEM Y

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